



EECS 841:

Lecture 12 (case study: face recognition)

Instructor: Sumaiya Shomaji,
Assistant Professor,
Dept. of EECS, University of Kansas
Email: shomaji@ku.edu
Office: 2020 Eaton Hall and 251 Nichols Hall

Slide Credit: Materials are used from "Introduction to Biometrics" text authors Arun Ross and Anil Jain as well as Dr. Walter Scheirer, Dr. Damon Woodard, and Ming-Hsuan Yang

Face: Useful Tool

- **Human face images are useful for many things:**
 - Person recognition
 - Gender, age, ethnicity, and emotional state of person (affect)
- **Face recognition has been studied extensively in computer vision**
- **Face is important biometric modality in many applications**
 - Law enforcement
 - Human-computer interaction
 - Surveillance

Face Recognition: The Definition

- Defined as establishing a person's identity based on facial characteristics
- In its simplest form, the problem of face recognition involves compare two face images and determining if they are of the same person.
- Humans are good at determining the similarity between two face images acquired under diverse conditions, while automated face recognition has several challenges.

Psychology of Face Recognition

If shown the cartoons, can you guess them?!



Albert Einstein



Bill Gates



Jim Carrey



Taylor Swift

- Humans can recognize caricatures and faces in cartoons
- Certain facial features are salient for each individual

www.magixl.com
www.pritchettcartoons.com
www.interchile.com/benjamin

The Truth: Overcome Face Variations to Recognize Faces- “The Challenge”

- **Face images of a person may have variations in:**
 - Changes in appearance due to facial hair, make-up, or accessories (glasses, piercings)
 - Age, pose, illumination, and expressions
- **The specific neural and cognitive processes involved in humans for face recognition are still not completely known.**

We still love “Face”!

Advantages of Using Face

- **Face can be captured at a longer standoff distance using non-contact sensors**
- **Face conveys identity and the emotion of a person as well as biographic information**
- **Large legacy face databases are available which can allow for the analysis of individuality and scalability**
- **Higher user acceptance**
 - People are more willing to share images of their face compared to other biometric traits such as fingerprint and iris.

How Did You Recognize

Ans: Based on the higher-level characteristics!

- Based on the observations that humans can recognize caricatures and cartoon faces, it is inferred that humans perceive the face based on certain higher-level characteristics.
- Studies using advanced brain imaging techniques such as functional magnetic resonance image (fMRI) are expected to reveal the precise face processing mechanism in the human brain.



Psychology of Face
recognition!

Facial Features: 3 level

- **Level 1 Details**
 - Consist of gross facial characteristics that are easily observable
- **Examples**
 - The general geometry of the face and global skin color
- **Such features can be used to discriminate between**
 - Faces exhibiting male and female characteristics
 - Faces of different races
 - A short round face and an elongated thin face
- **These features can be extracted even from low resolution face images, e.g. < 30 interpupillary distance (IPD)**

Facial Features: 3 level

- **Level 2 Details**

- Consists of localized face information such as the structure of the face components, the relationship between facial components, and the precise shape of the face
- These features are essential for accurate face recognition, and they require a higher resolution face image (30 – 75 IPD). The characteristics of local regions of the face can be represented using geometric or texture descriptors.

Facial Features: 3 level

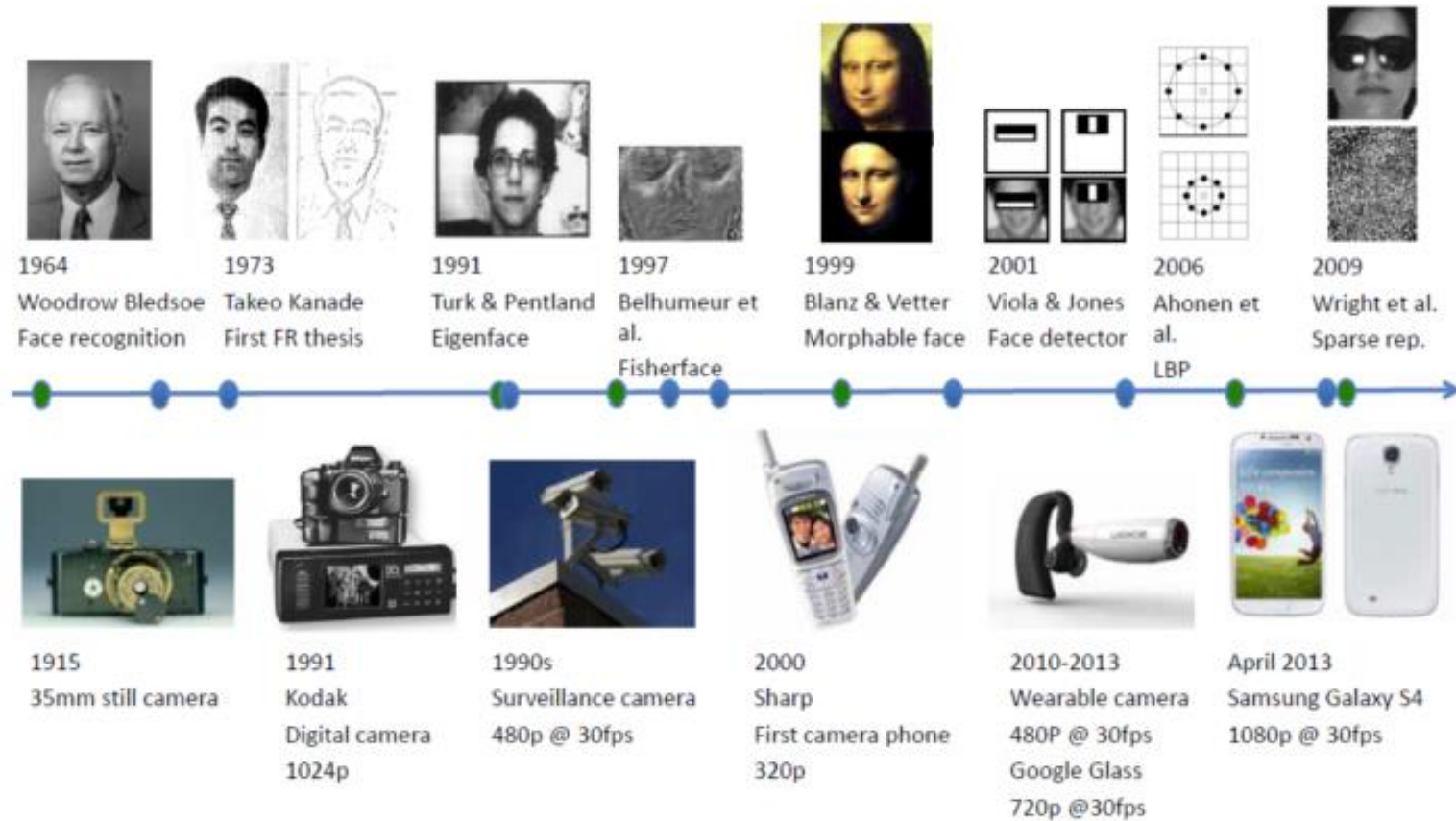
- **Level 3 Details**

- Consist of unstructured, micro level features on the face, which includes scars, freckles, skin discoloration, and moles
- One challenging face recognition problem where level 3 details may be critical is the discrimination of identical twins.

Part II: Face recognition

Not a very new technology!

Face Recognition Milestones



Bledsoe, W. W. 1964. The Model Method in Facial Recognition, TR PRI 15, Panoramic Research, Inc., California.
 Takeo Kanade, Picture Processing System by Computer Complex and Recognition of Human Faces, Kyoto Univ., 1973.
 M. Turk and A. Pentland, Eigenfaces for recognition, *Journal of Cognitive Neuroscience* 3 (1): 71–86, 1991.
 Belhumeur, P.N. et al., Eigenfaces vs. Fisherfaces: recognition using class specific linear projection, *PAMI*, 19-7, 1997.
 V. Blanz and T. Vetter, A morphable model for the synthesis of 3D faces, *SIGGRAPH* 1999.
http://commons.wikimedia.org/wiki/File:Three_Surveillance_cameras.jpg
<http://photoshoto.com/camera-history-timeline/>
 Viola, Jones: Robust Real-time Object Detection, *ICV* 2001.
 Ahonen et al. Face Description with Local Binary Patterns: Application to Face Recognition, *PAMI*, 2006.
 J. Wright et al. Robust Face Recognition via Sparse Representation, *PAMI*, 31-2, 2009.
<http://static7.businessinsider.com/images/4d313ea7cadcb7033010000/lookie-video-camera.jpg>
<http://www.techinasia.com/samsung-galaxy-s4-infographic/>

Face Recognition

- Face is the most common biometric characteristic used by humans for recognition
- Applications range from static, mug-shot verification to a dynamic, uncontrolled face identification in a cluttered background

More Challenges

Inter-class Similarity

- Different persons may have very similar appearance



www.marykateandashley.com

Twins



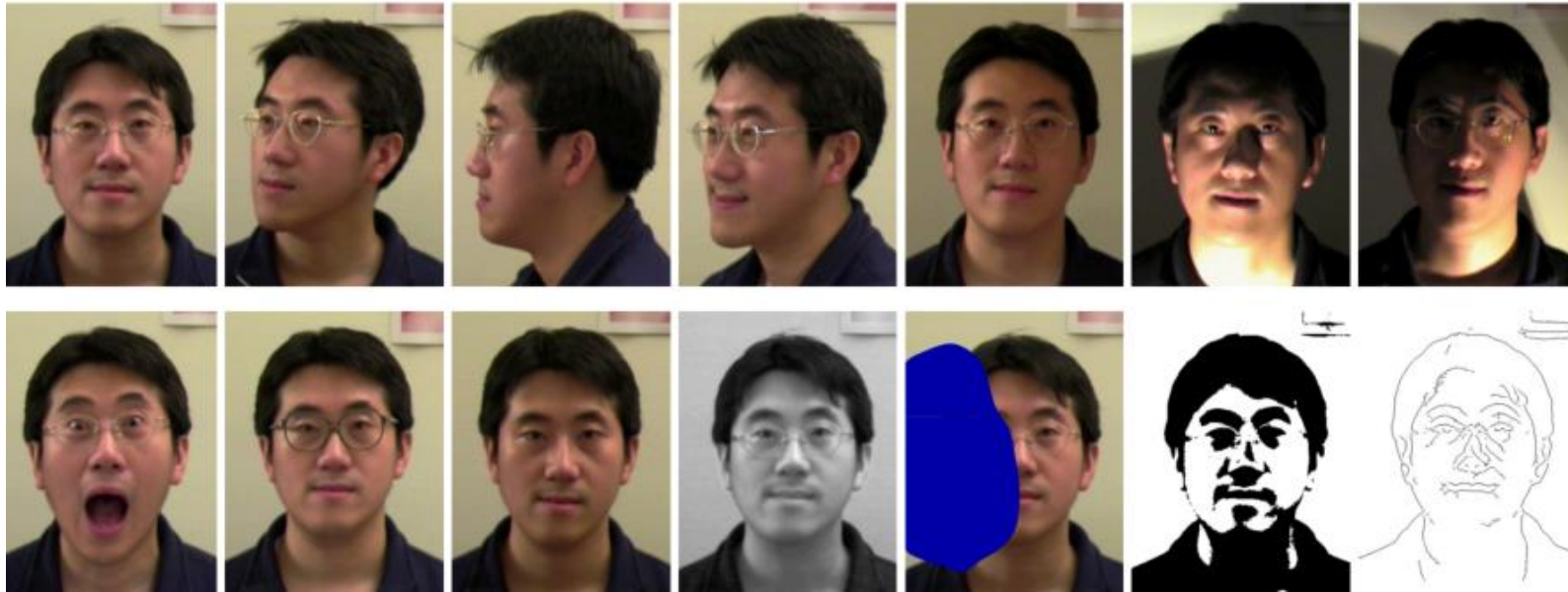
news.bbc.co.uk/1/hi/english/in_depth/americas/2000/us_elections

Father and son

More Challenges

Intra-class Variability

- Faces with intra-subject variations in pose, illumination, expression, accessories, color, occlusions, and brightness



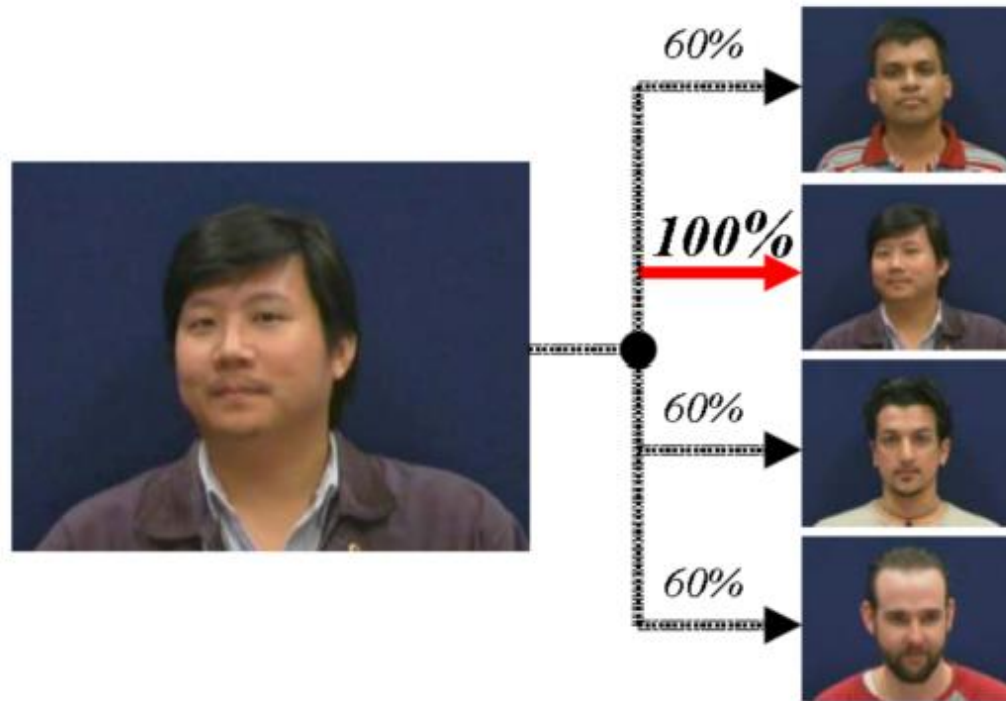
Face Recognition in the Literature

- **Appearance-based methods:** generate a compact representations of the entire face region in the acquired image by mapping the high dimensional face image to a lower dimensional subspace.
- **Texture-based methods:** use textural features to represent and detect facial patterns
- **Model-based methods:** attempt to build a 2D or 3D face models that facilitate matching of faces images in the presence of pose changes.

Appearance Based Recognition

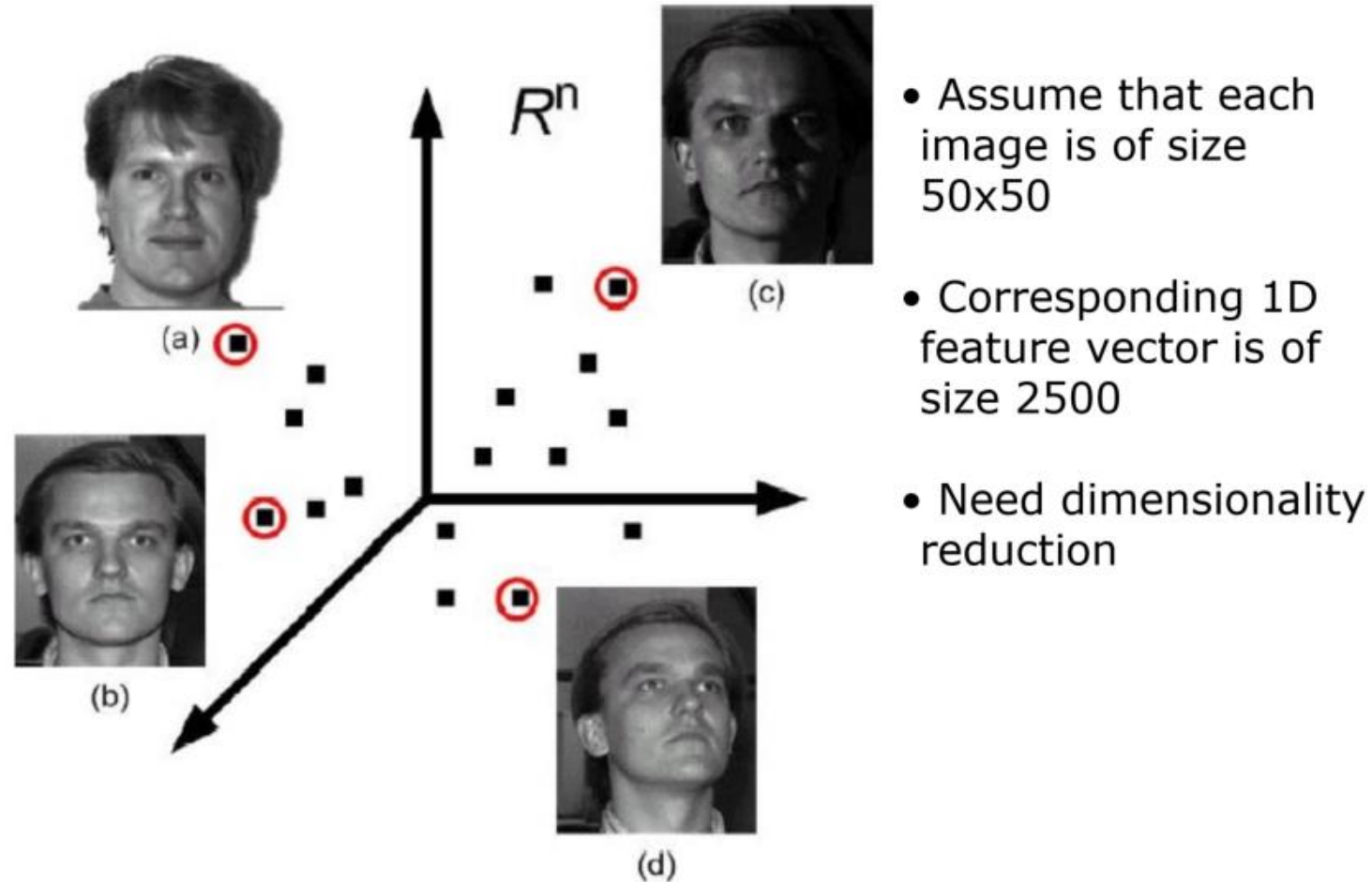
- **Appearance-based methods:** generate a compact representations of the entire face region in the acquired image by mapping the high dimensional face image to a lower dimensional subspace.
- **Texture-based methods:** use textural features to represent and detect facial patterns
- **Model-based methods:** attempt to build a 2D or 3D face models that facilitate matching of faces images in the presence of pose changes.

Appearance Based Recognition: Beginning



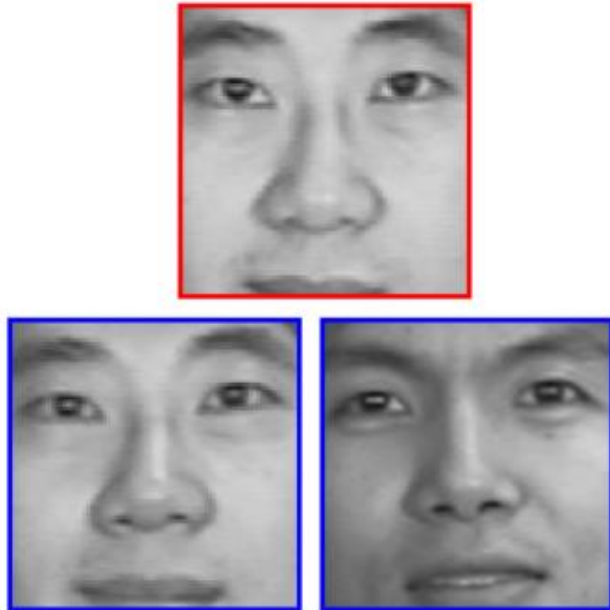
Two images are superimposed and the correlation between corresponding pixels is computed for different alignments.

Appearance Based Recognition: Challenge



Appearance Based Recognition:

- Derive features based on the pixel values of the face image



Example:

- Convert 2D pixel intensity values of the image into a 1D feature vector
- Transform this feature vector to a low-dimensional salient representation
- Face recognition involves comparing two such representations (feature vectors)

Texture based Facial Recognition

Local Binary Patterns (LBP)

Motivation:

2-D surface texture is a two-dimensional phenomenon characterized by:

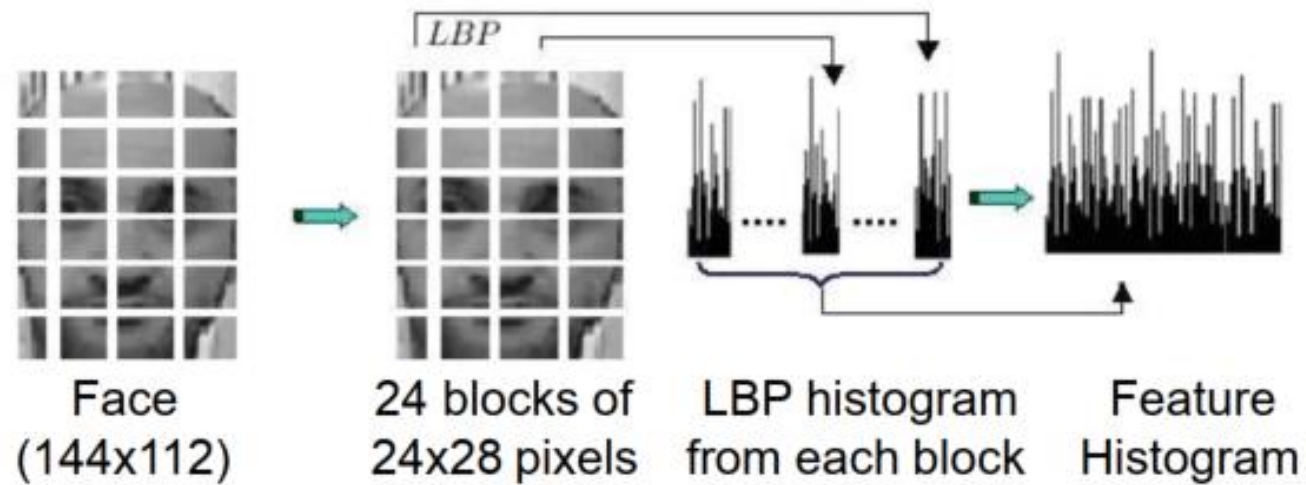
1. Spatial Structure (pattern)
2. Contrast (amount of texture)



Adapted from notes by M. Pietikäinen and G. Zhao

LBP for Face Recognition with Histogram # 1

T.Ahonen et al. IEEE T-PAMI 2006



SVM or other supervised learning for classification

LBP for Face Recognition with Binary Feature Descriptor # 2

3x3 neighborhood

example

6	5	2
7	6	1
9	8	7

thresholded

1	0	0
1		0
1	1	1

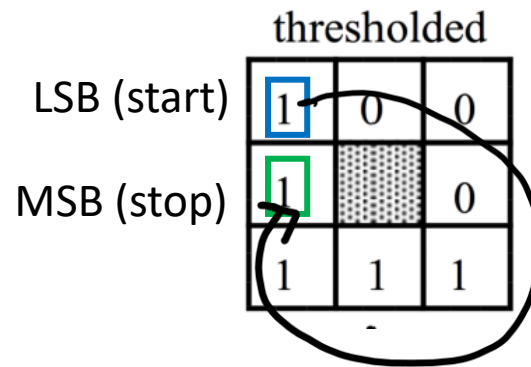
weights

1	2	4
128		8
64	32	16

Binary Pattern is created from the thresholded matrix by accessing and concatenating each binary values (start accessing clock-wise from the top left corner and consider that as LSB)

Binary Pattern = ?

Binary Pattern = 11110001



LBP for the  cell= $\textcolor{blue}{1} \times 2^0 + 0 \times 2^1 + 0 \times 2^2 + 0 \times 2^3 + 1 \times 2^4 + 1 \times 2^5 + 1 \times 2^6 + \textcolor{green}{1} \times 2^7$

$$= 1 + 16 + 32 + 64 + 128 = 241$$

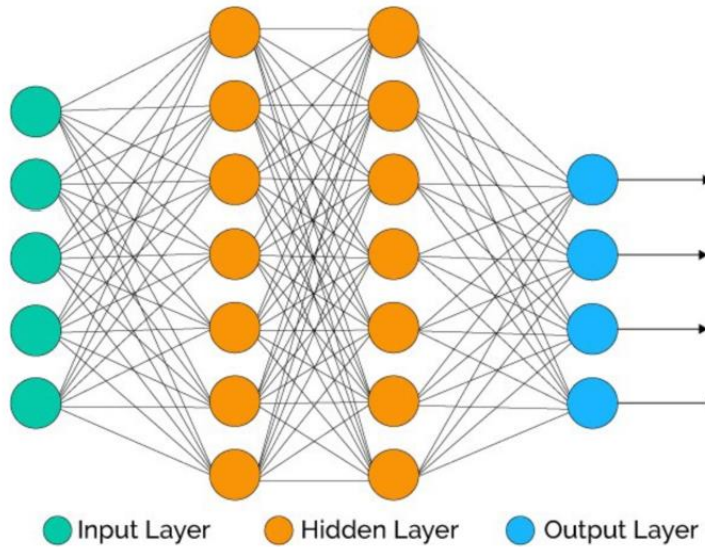
Model Based Recognition

Appearance Based Recognition

- **Appearance-based methods:** generate a compact representations of the entire face region in the acquired image by mapping the high dimensional face image to a lower dimensional subspace.
- **Texture-based methods:** use textural features to represent and detect facial patterns
- **Model-based methods:** attempt to build a 2D or 3D face models that facilitate matching of faces images in the presence of pose changes.

Model based Method

A simple neural network

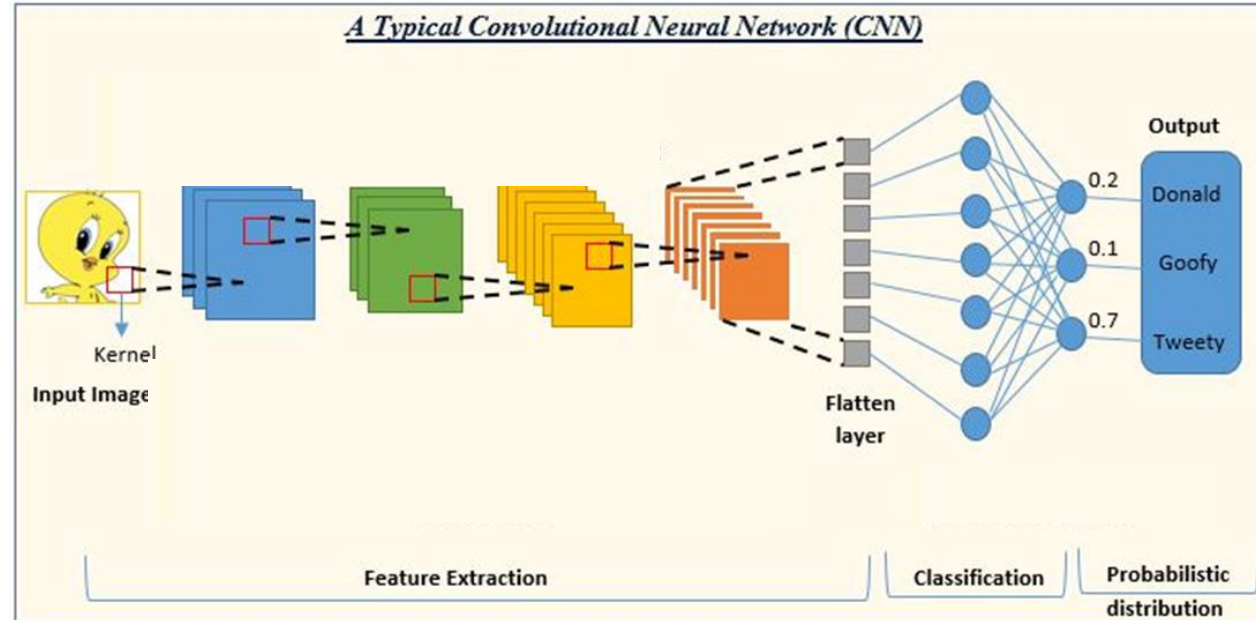


Model based Method: CNN

Convolutional Neural Network (CNN):

- Deep Learning algorithm which takes in an input image, assign importance (learnable weights and biases) to various aspects/objects in the image and based on that differentiates one from the other.

2 Parts: Feature extraction and Classification



Feed-forward: algorithm to calculate output vector from input vector

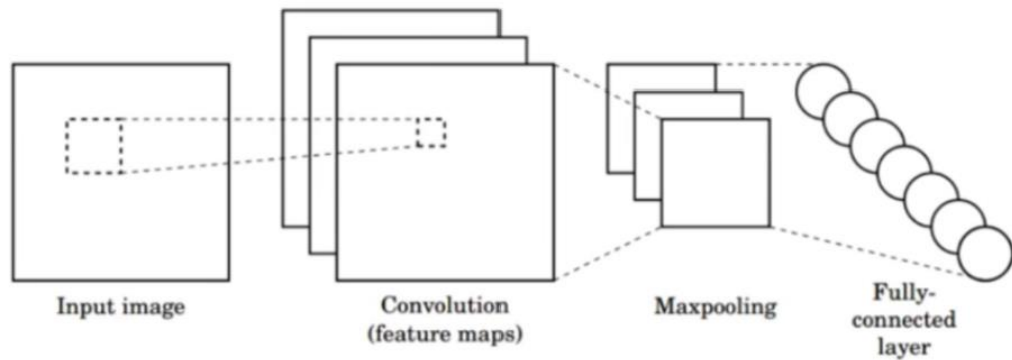
Backpropagation: algorithm to train (adjust weight) of neural network. Input for backpropagation is output_vector, target_output_vector, and adjusted_weight_vector.

Model based Method: CNN

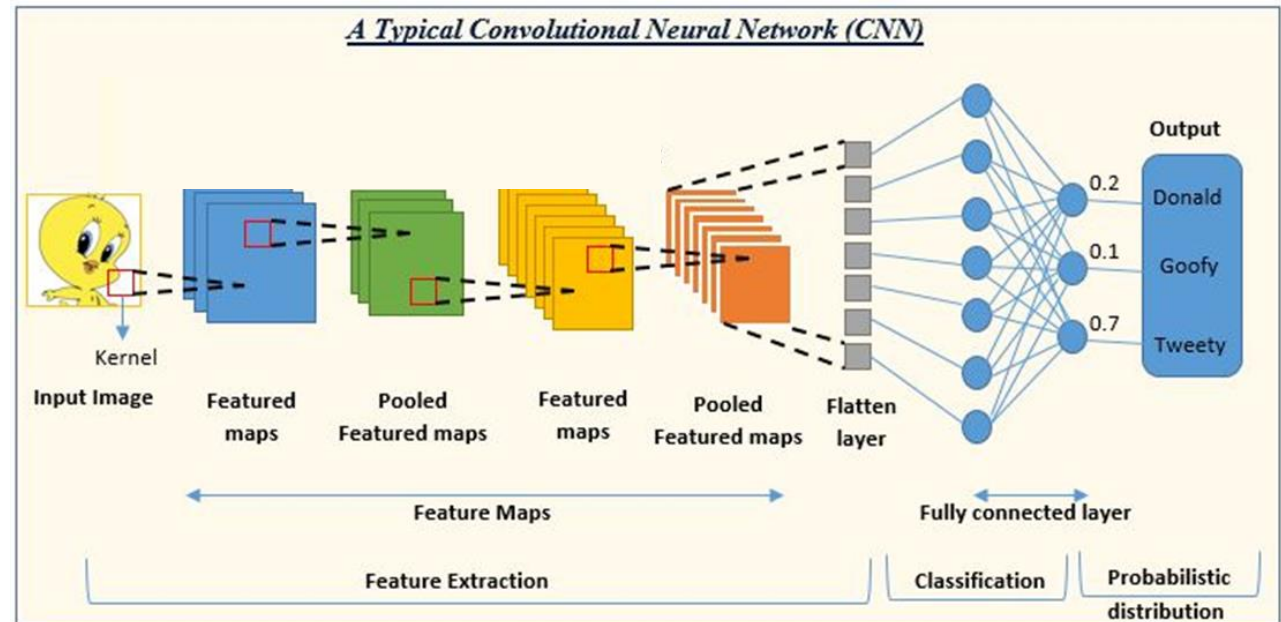
Convolutional Neural Network (CNN):

- Deep Learning algorithm which takes in an input image, assign importance (learnable weights and biases) to various aspects/objects in the image and based on that differentiates one from the other.

A simple CNN example:



2 Parts: Feature extraction and Classification



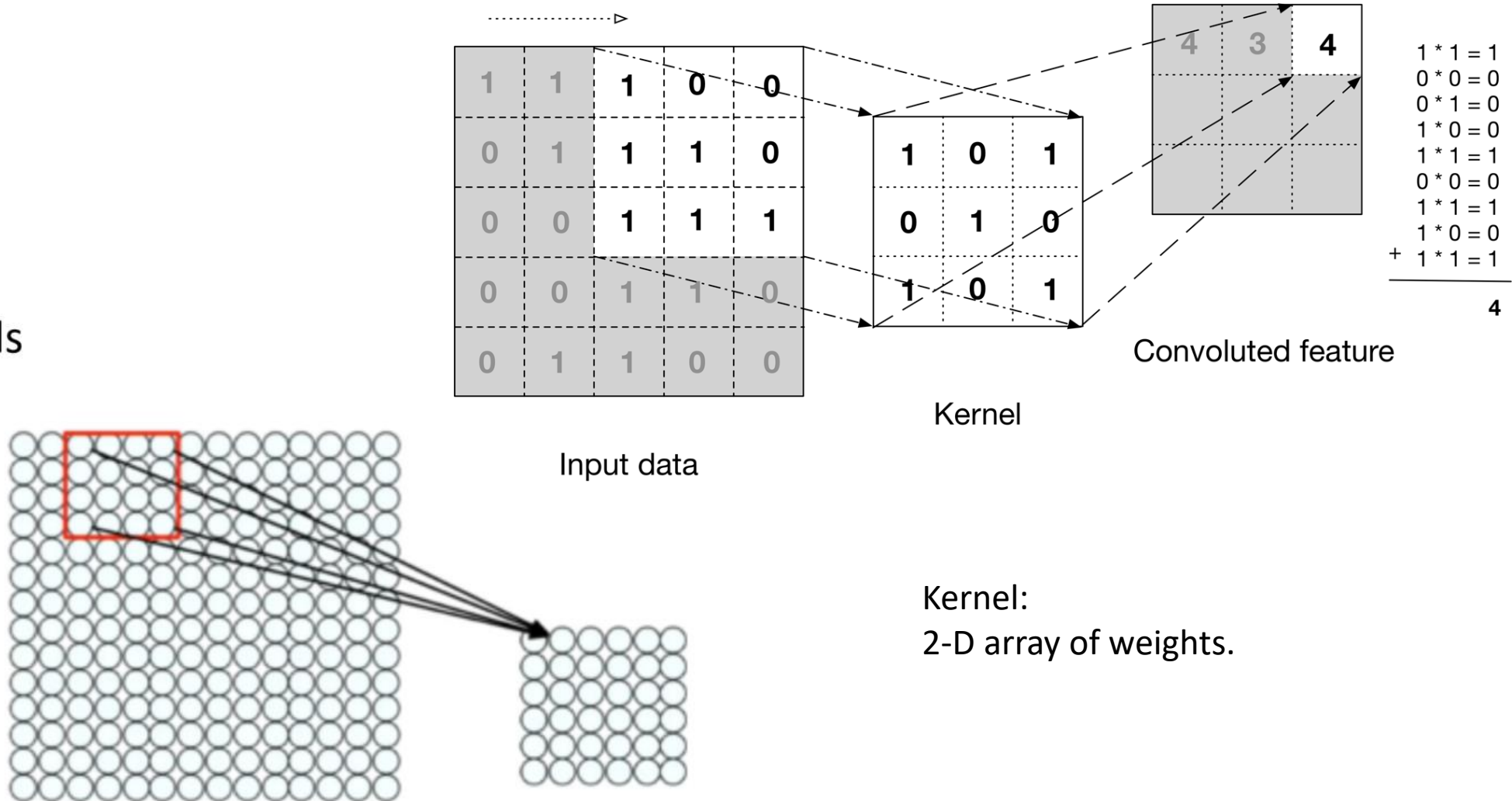
Feed-forward: algorithm to calculate output vector from input vector

Backpropagation: algorithm to train (adjust weight) of neural network. Input for backpropagation is output_vector, target_output_vector, and adjusted_weight_vector.

Feature Learning/Feature Extraction

Convolution

- Kernel size
- Stride
- Number of kernels

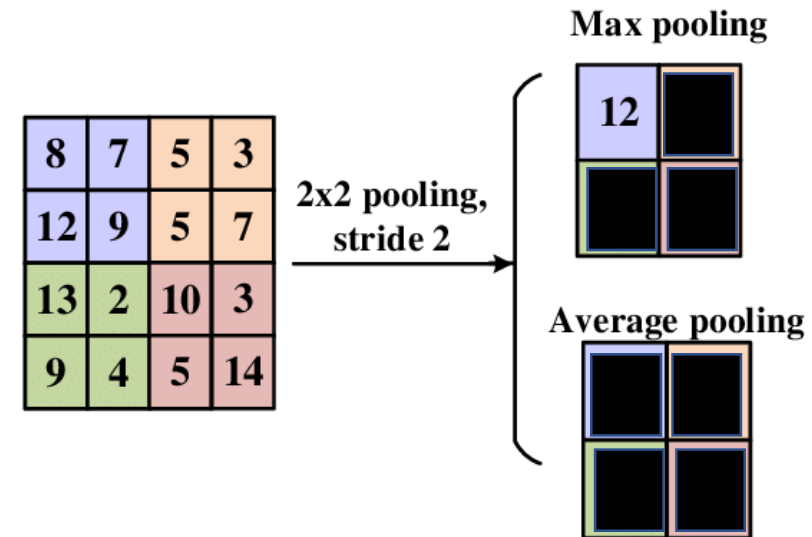


Model based Method

Other parameters

A Pooling layer is added after the Convolutional layer(s). It **downsamples the output of the Convolutional layers by sliding the filter of some size with some stride size and calculating the maximum or average of the input**

- Pooling Layers
- Non-linear activations
- Fully-connected Layer

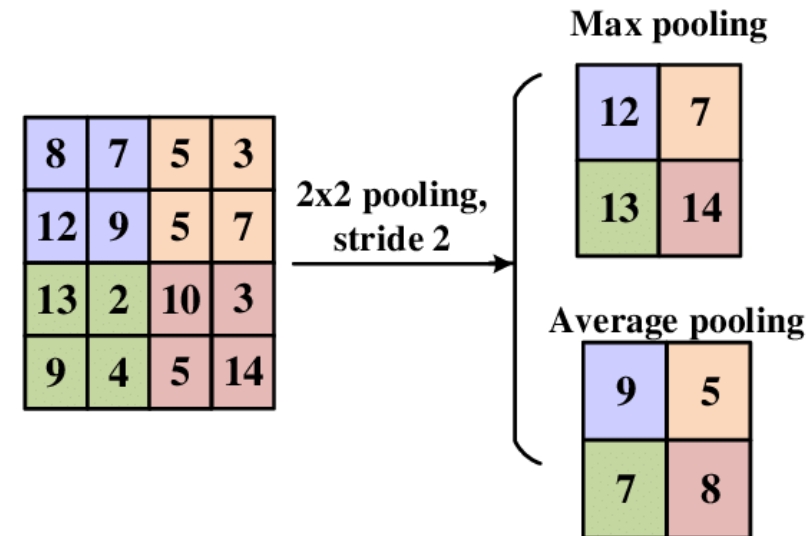


Model based Method

Other parameters

A Pooling layer is added after the Convolutional layer(s). It **downsamples the output of the Convolutional layers by sliding the filter of some size with some stride size and calculating the maximum or average of the input**

- Pooling Layers
- Non-linear activations
- Fully-connected Layer

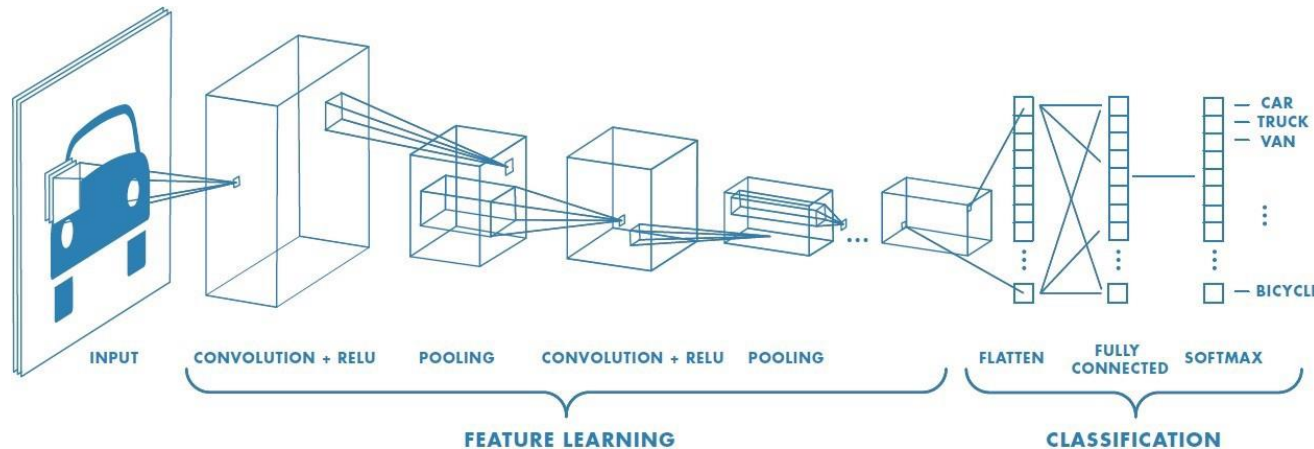
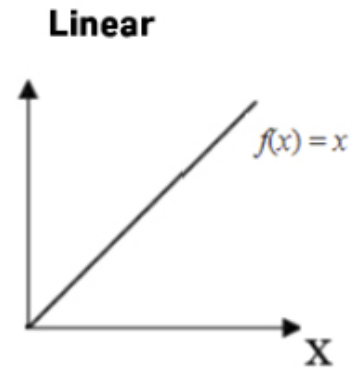
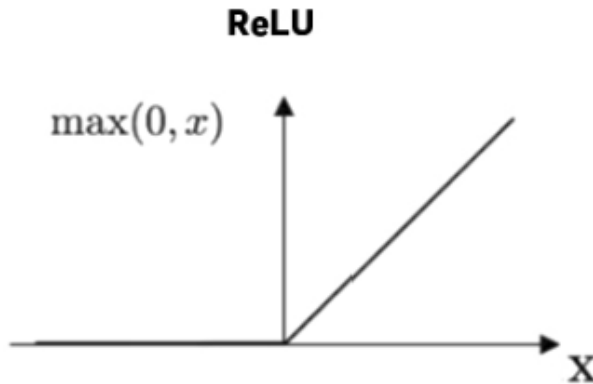


Model based Method

Other parameters

- Pooling Layers
- Non-linear activations
- Fully-connected Layer

An activation function in a neural network **defines how the weighted sum of the input is transformed into an output from a node or nodes in a layer of the network** → transform function

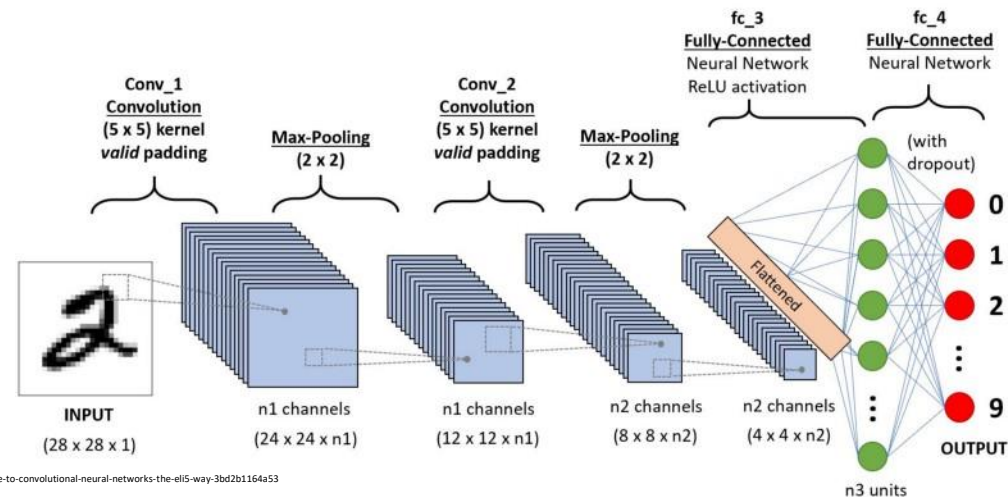
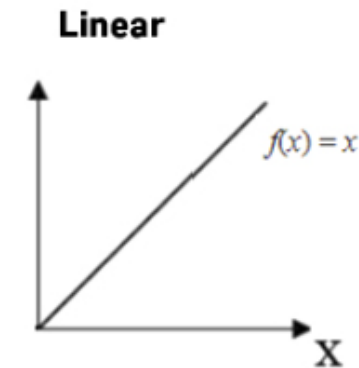
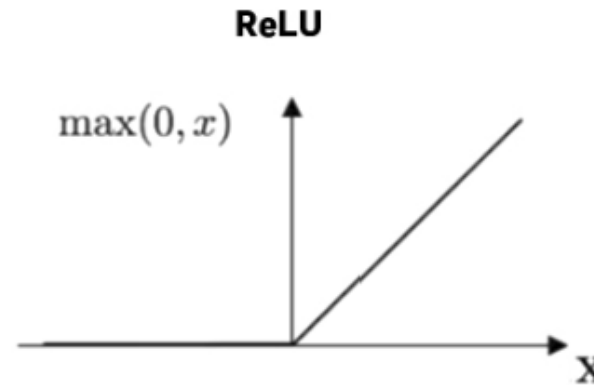


Model based Method

Other parameters

- Pooling Layers
- Non-linear activations
- Fully-connected Layer

An activation function in a neural network **defines how the weighted sum of the input is transformed into an output from a node or nodes in a layer of the network** → transform function



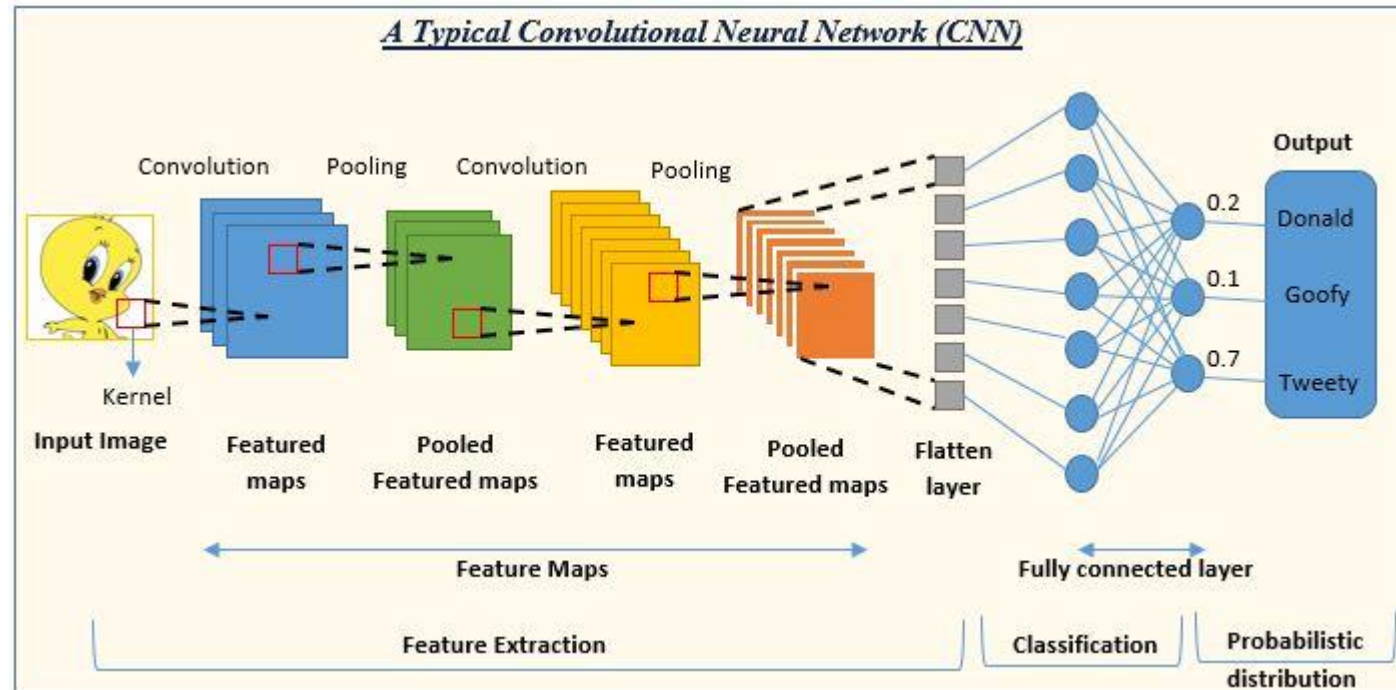
Model based Method

Other parameters

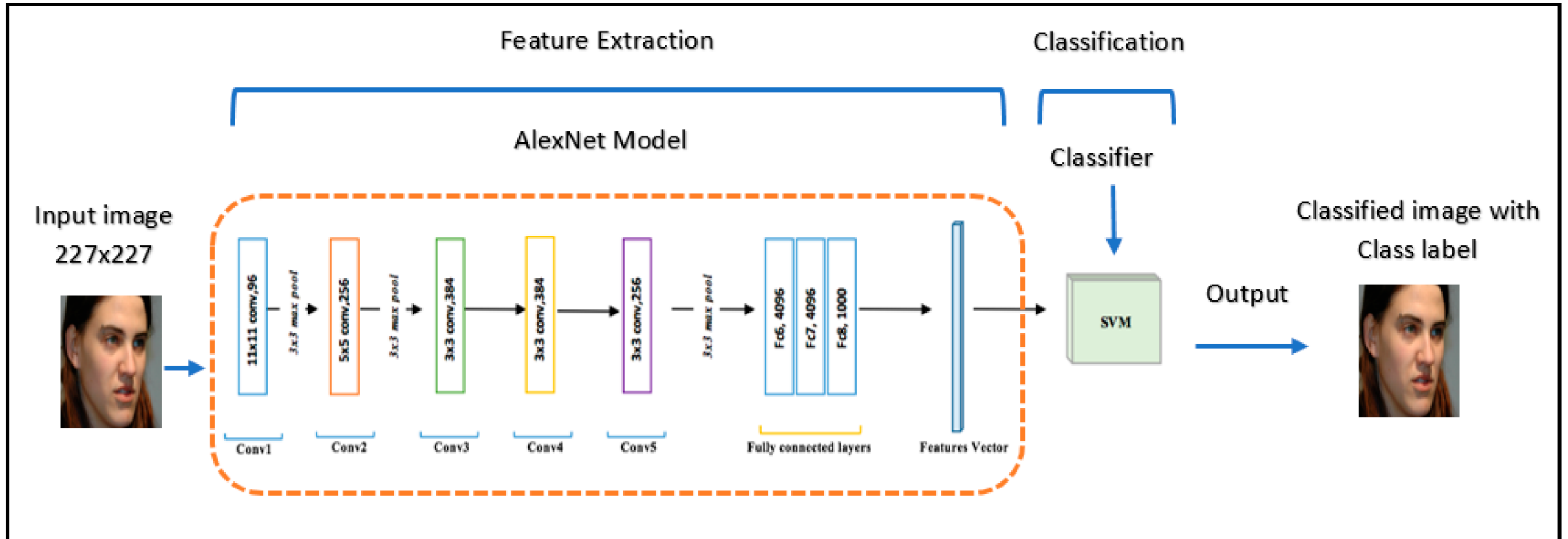
- Pooling Layers
- Non-linear activations
- Fully-connected Layer

Fully Connected Layer is simply, **feed forward neural networks**. They form the last few layers in the network. The input to the fully connected layer is the output from the final Pooling or Convolutional Layer, which is flattened and then fed into the fully connected layer.

Dense layer, also called fully-connected layer, refers to the layer whose inside neurons connect to every neuron in the preceding layer. They are meant for classification.



Face Recognition with CNN



Popular Architectures

Model	Size	Top-1 Accuracy	Top-5 Accuracy	Parameters	Depth
VGG16	528 MB	0.713	0.901	138,357,544	23
InceptionV3	92 MB	0.779	0.937	23,851,784	159
ResNet50	98 MB	0.749	0.921	25,636,712	-
Xception	88 MB	0.790	0.945	22,910,480	126
InceptionResNetV2	215 MB	0.803	0.953	55,873,736	572
ResNeXt50	96 MB	0.777	0.938	25,097,128	-

Popular Architectures with Dataset

Deep Face Recognition: A Survey

Mei Wang, Weihong Deng

Method	Public. Time	Loss	Architecture	Training Set	Accuracy $\pm$ Std(%)
DeepFace [20]	2014	softmax	Alexnet	Facebook (4.4M,4K)	97.35 $\pm$ 0.25
DeepID2 [21]	2014	contrastive loss	Alexnet	CelebFaces+ (0.2M,10K)	99.15 $\pm$ 0.13
DeepID3 [36]	2015	contrastive loss	VGGNet-10	CelebFaces+ (0.2M,10K)	99.53 $\pm$ 0.10
FaceNet [38]	2015	triplet loss	GoogleNet-24	Google (500M,10M)	99.63 $\pm$ 0.09
Baidu [58]	2015	triplet loss	CNN-9	Baidu (1.2M,18K)	99.77
VGGface [37]	2015	triplet loss	VGGNet-16	VGGface (2.6M,2.6K)	98.95
light-CNN [85]	2015	softmax	light CNN	MS-Celeb-1M (8.4M,100K)	98.8
Center Loss [101]	2016	center loss	Lenet+-7	CASIA-WebFace, CACD2000, Celebrity+ (0.7M,17K)	99.28
L-softmax [104]	2016	L-softmax	VGGNet-18	CASIA-WebFace (0.49M,10K)	98.71
Range Loss [82]	2016	range loss	VGGNet-16	MS-Celeb-1M, CASIA-WebFace (5M,100K)	99.52
L2-softmax [109]	2017	L2-softmax	ResNet-101	MS-Celeb-1M (3.7M,58K)	99.78
Normface [110]	2017	contrastive loss	ResNet-28	CASIA-WebFace (0.49M,10K)	99.19
CoCo loss [112]	2017	CoCo loss	-	MS-Celeb-1M (3M,80K)	99.86
vMF loss [115]	2017	vMF loss	ResNet-27	MS-Celeb-1M (4.6M,60K)	99.58
Marginal Loss [116]	2017	marginal loss	ResNet-27	MS-Celeb-1M (4M,80K)	99.48
SphereFace [84]	2017	A-softmax	ResNet-64	CASIA-WebFace (0.49M,10K)	99.42
CCL [113]	2018	center invariant loss	ResNet-27	CASIA-WebFace (0.49M,10K)	99.12
AMS loss [105]	2018	AMS loss	ResNet-20	CASIA-WebFace (0.49M,10K)	99.12
Cosface [107]	2018	cosface	ResNet-64	CASIA-WebFace (0.49M,10K)	99.33
Arcface [106]	2018	arcface	ResNet-100	MS-Celeb-1M (3.8M,85K)	99.83
Ring loss [117]	2018	Ring loss	ResNet-64	MS-Celeb-1M (3.5M,31K)	99.50

Suggested variations in dataset:

Pose Variations

Cross-Age FR

Illumination Changes

Facial Expression Variations

Facial Occlusions

Facial Makeup

Mixed Variations

<https://arxiv.org/pdf/1804.06655.pdf>

Face Recognition with CNN

Advantages:

- Automatically detects the important features without any human supervision
- Earns distinctive features for each class by itself
- Allows the model to learn position and scale of faces in an image

Challenges:

- Inherent bias in large scale dataset
- Optimizing architecture
- Training time

Research Direction

- Robustly recognizing faces
 - Small and/or noisy images
 - Images acquired years apart
 - Outdoor acquisition: lighting, pose
- Change in 3D head pose and tilt
- Varying illumination
- Different facial expressions
- Effects of aging
- Facial accessories
- Partial or occluded face in surveillance systems
- What is the principal and optimal way to combine local features and global features

---THE END---